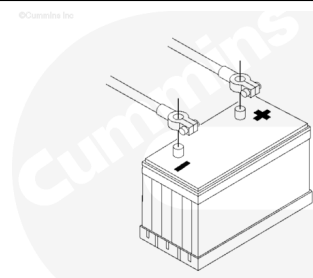


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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Disconnect the batteries.

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



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Note : The minimum clearance required from the front gear housing face to any obstruction in the camshaft area for camshaft removal is 81.28 cm [32 in]. If the required amount of clearance is not available, the engine must be removed from the vehicle. Refer to Procedure 000-001 (</qs3/pubsys2/xml/en/procedures/40/40-000-001.html>).

- Remove the engine water pump (cooling fan) drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html>)
- If required, remove the fan hub. Refer to Procedure 008-039 (</qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html>)
- Remove the vibration damper. If equipped with a viscous damper, refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html>). If equipped with a rubber damper, refer to

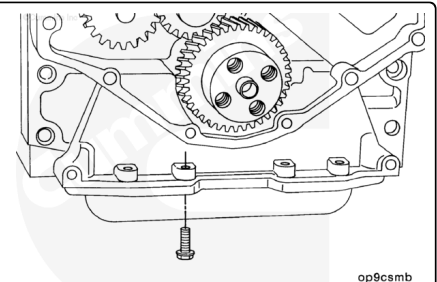
Procedure 001-051

(/qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html)

- Remove the front gear cover. Refer to Procedure 001-031 (/qs3/pubsys2/xml/en/procedures/40/40-001-031-tr.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html)
- Remove the rocker levers. Refer to Procedure 003-008 (/qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html)
- Remove the push tubes. Refer to Procedure 004-014 (/qs3/pubsys2/xml/en/procedures/40/40-004-014-tr.html)
- Remove the fuel transfer pump. Refer to Procedure 005-045 (/qs3/pubsys2/xml/en/procedures/40/40-005-045-tr.html)
- Use tappet replacement kit, Part Number 3822513, to raise the tappets. Refer to Procedure 004-015 (/qs3/pubsys2/xml/en/procedures/40/40-004-015-tr.html)
- Remove the camshaft. Refer to Procedure 001-008 (/qs3/pubsys2/xml/en/procedures/40/40-001-008-tr.html)
- Remove the fuel injection pump. For rotary fuel injection pumps, refer to Procedure 005-014 (/qs3/pubsys2/xml/en/procedures/40/40-005-014-tr.html). For inline fuel injection pumps, refer to Procedure 005-012 (/qs3/pubsys2/xml/en/procedures/40/40-005-012-tr.html)
- Remove or disconnect driven accessories (i.e., hydraulic pump).

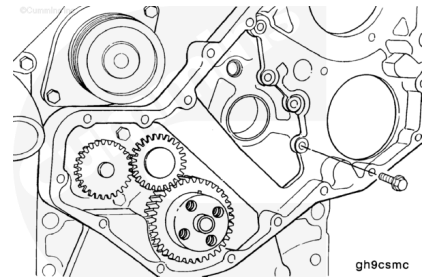
Remove

Remove the four front oil pan capscrews.

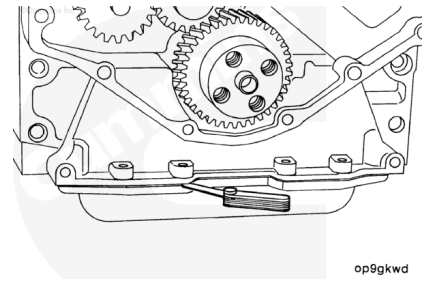


Remove the gear housing capscrews.

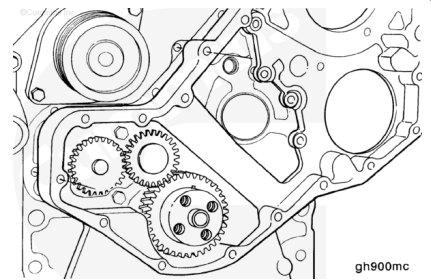
Note the location of the gear housing capscrews as they are removed. Some of the capscrews are an internal torx fastener and **must** be installed in the same location as removed to ensure proper clearance.



Using a feeler gauge, separate the lubricating oil pan gasket from the gear housing.



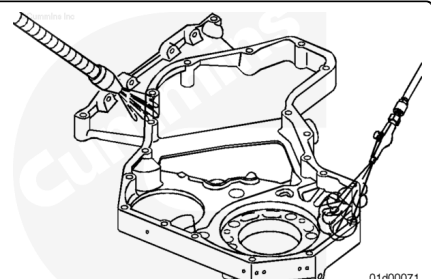
Use a plastic hammer to loosen the front gear housing.
Remove the gear housing.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



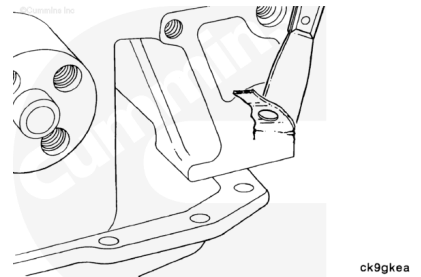
⚠ WARNING ⚠

Compressed air used for cleaning should not exceed 207 kPa [30 psi]. Wear protective clothing and goggles to avoid personal injury.

Use solvent to clean the gear housing.

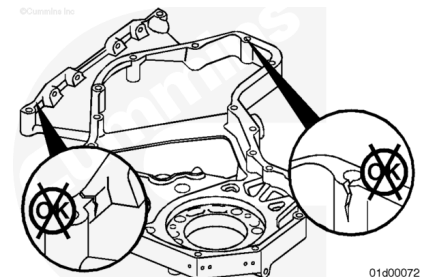
Dry the housing with compressed air.

Clean all gasket material from the cylinder block.

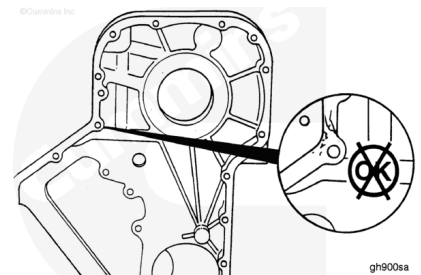


Inspect the gear housing for cracks or damage.

If any damage is found, replace the gear housing.

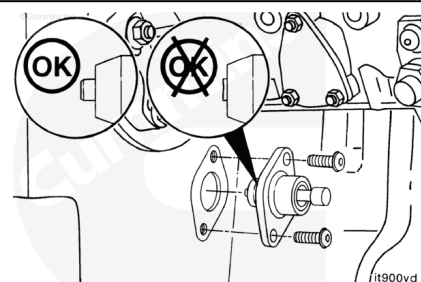


Note : If the front gear housing is cracked at the accessory drive hole or fuel injection pump mounting location, double check the mounting brackets, fasteners and component installation practices.

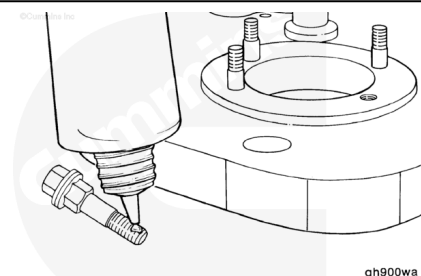


Inspect the timing pin housing and pin for damage.

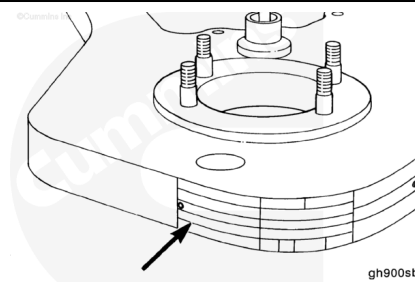
Note : Do not remove the timing pin assembly unless it is damaged or leaking, or the gear housing is being replaced. Refer to Procedure 001-049 (</qs3/pubsys2/xml/en/procedures/40/40-001-049.html>).



If the fuel injection pump studs are damaged or being installed in a new housing, coat the threads with Loctite 242. Use two nuts locked together to remove and install the studs.



If the gear housing is being replaced, remove the engine dataplate and install it on the new gear housing; refer to Procedure 001-057 (/qs3/pubsys2/xml/en/procedures/40/40-001-057.html).



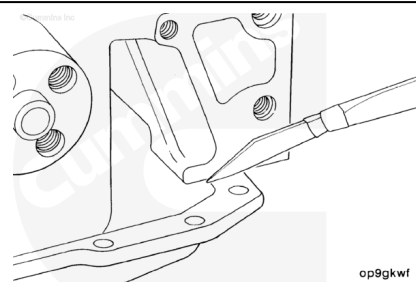
Install

For engines equipped with a paper gasket style oil pan, inspect the oil pan gasket for damage.

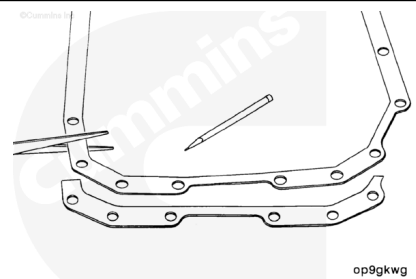
Note : If the engine is equipped with a suspended oil pan with a rubber oil pan gasket and the gasket is found to be damaged, the oil pan must be removed and the entire oil pan gasket replaced. Refer to Procedure 007-025 (/qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html). If the engine used sealant only between the gear housing and the oil pan, remove the old sealant and apply new sealant, Part Number 3164070, to the oil pan mounting flange.

If the pan gasket is torn, it can be repaired.

Cut the torn gasket off even with the front of the cylinder block.



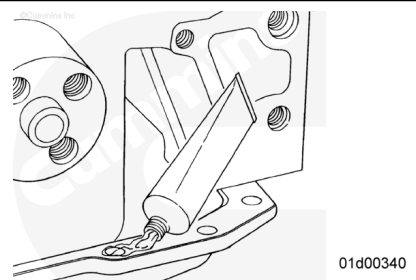
Using the old gasket as a pattern, cut the front section of a new gasket to the same size.



Note : The gear housing must be installed within 10 minutes of applying the sealant.

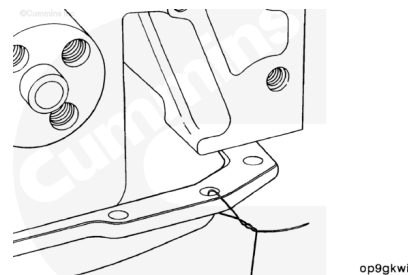
Clean the sealing surfaces.

Coat the new gasket on both sides with sealant, Part Number 3164067.



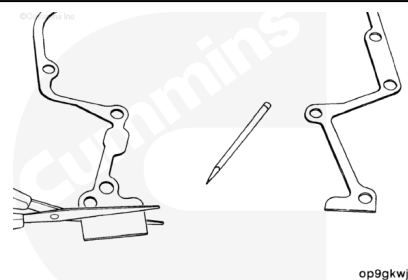
Be sure there is a bead of sealant at the intersecting joint of the cylinder block, oil pan, and gear housing.

Use common thread or very fine wire to hold the new gasket splice in position as illustrated.



Check the fit of the new gear housing to cylinder block gasket. It may be necessary to trim the bottom edges of the gasket before installation.

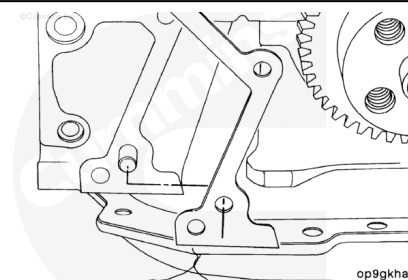
Note : When properly trimmed, the gear housing gasket must be even with the lubricating oil pan gasket when installed.



Position the gasket on the alignment dowels.

Use guide pins, M8 x 1.25 x 50, to assist in aligning the gasket and gear housing. Make sure to remove the guide pins after alignment.

Apply a bead of sealant, Part Number 3164067, at the intersecting joint of the cylinder block, oil pan, and gear housing.



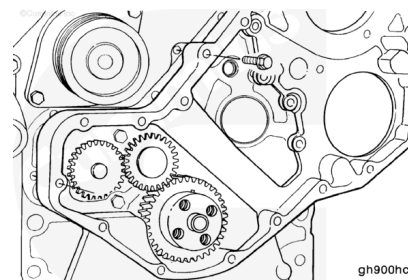
Carefully install the gear housing.

Make sure the gasket is still in place.

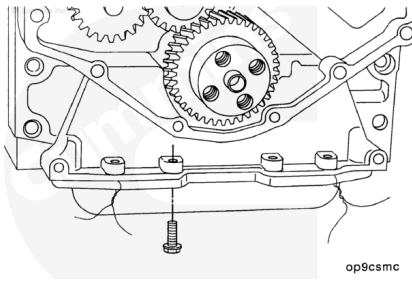
Install and tighten the mounting capscrews.

Torque Value: 24 n•m [18 ft-lb]

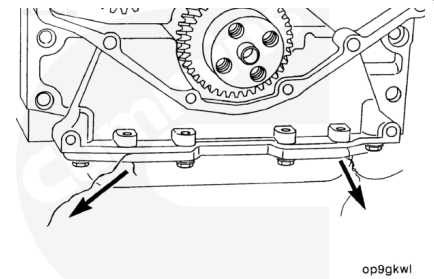
Note : If a new front gear housing and/or timing pin assembly is installed, the timing pin assembly must be accurately located. Refer to Procedure 001-049 (/qs3/pubsys2/xml/en/procedures/40/40-001-049.html).



Start the oil pan capscrews into the holes **not** being used to tie the gasket in place.



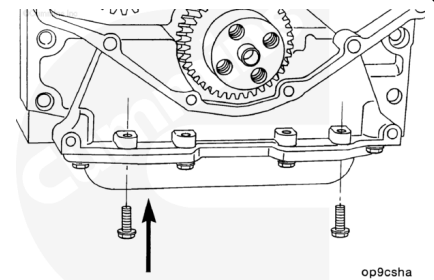
Remove the thread or wire holding the gasket in place.



Install the remaining two capscrews.

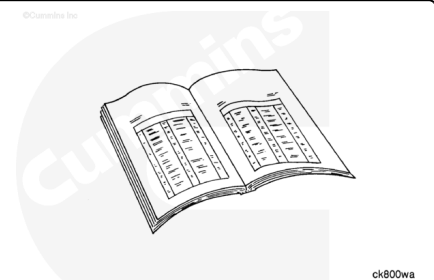
Tighten the capscrews.

Torque Value: 24 n•m [18 ft-lb]



Finishing Steps

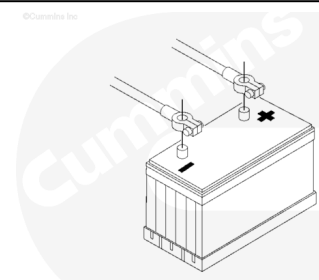
- Install the camshaft. Refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/40/40-001-008-tr.html>)
- Release the tappets. Refer to Procedure 004-015 (</qs3/pubsys2/xml/en/procedures/40/40-004-015.html>)
- Install the push rods. Refer to Procedure 004-014 (</qs3/pubsys2/xml/en/procedures/40/40-004-014-tr.html>)
- Install the rocker levers. Refer to Procedure 003-008 (</qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html>)
- Adjust the valve lash. Refer to Procedure 003-004 (</qs3/pubsys2/xml/en/procedures/40/40-003-004-tr.html>)



- Install the rocker lever cover. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html>)
- Install the fuel transfer pump. Refer to Procedure 005-045 (</qs3/pubsys2/xml/en/procedures/40/40-005-045-tr.html>)
- Install the fuel injection pump. For rotary fuel injection pumps, refer to Procedure 005-014 (</qs3/pubsys2/xml/en/procedures/40/40-005-014-tr.html>). For inline fuel injection pumps, refer to Procedure 005-012 (</qs3/pubsys2/xml/en/procedures/40/40-005-012-tr.html>).
- Install or connect driven accessories (i.e., hydraulic pump)
- Install the front gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/40/40-001-031-tr.html>)
- Install the vibration damper. If equipped with a viscous damper, refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html>). If equipped with a rubber damper, refer to Procedure 001-051 (</qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html>)
- If removed, install the fan hub. Refer to Procedure 008-039 (</qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html>)
- Install the engine water pump drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html>).

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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Connect the batteries.

Operate the engine at idle for 5 to 10 minutes and check for leaks or loose parts.

